Comprehensive Bulk RNA-seq Analysis Report

Hashimoto's Thyroiditis (HT) vs Non-HT: Transcriptomic Profiling, Pathway Enrichment & Drug Target Prioritization

Analyst: Biomni (Phylo) **Date:** 2026-04-13 **Comparison:** HT vs Non-HT (reference)
Pipeline: DESeq2 differential expression | ORA enrichment | CIBERSORTx deconvolution

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> ⚠️ **CRITICAL LIMITATIONS — READ BEFORE INTERPRETING ANY RESULT**
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> 1. **Sample size (n=2 per group):** This is a pilot-scale dataset. DESeq2 has severely limited statistical power with n=2. All findings are **hypothesis-generating only** and must be validated in a larger, independent cohort.
> 2. **Outlier sample (HRR568894, HT):** This HT sample clusters closer to Non-HT samples on PC2 (28% variance) and shows lower within-group correlation (r=0.946 with HRR568844) than expected. This within-group heterogeneity inflates variance estimates and may suppress or distort some DEG calls.
> 3. **Threshold discrepancy:** The pre-existing "Upregulated" sheet in the source file contains 1,502 genes (log2FC > 1, no padj filter). Applying the strict padj ≤ 0.05 filter reduces this to **986 upregulated** and **166 downregulated** genes. The ORA enrichment results were likely computed on the looser 1,502-gene list, creating a minor inconsistency between DEG counts and enrichment inputs.
> 4. **No KEGG pathways are significantly enriched in downregulated genes** after FDR correction (all padj > 0.13). Only GO Biological Process terms reach significance for the downregulated set.

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1. Data Audit & Overview

Dataset Summary

Parameter	Value
Comparison	HT vs Non-HT
Total samples	4 (n=2 HT: HRR568844, HRR568894; n=2 Non-HT: HRR568887, HRR568888)
Statistical method	DESeq2 (variance-stabilizing transformation for QC)
Total genes tested	16,163 protein-coding genes
Normalization	VST (Variance Stabilizing Transformation)

Files Identified

Folder	Contents
`4. Differential Gene Expression Reports`	DEG table (all results + filtered sheets), volcano plot, unsupervised heatmap
`5. Functional enrichment reports`	ORA results (GO BP/MF/CC + KEGG) for up- and downregulated genes
`6. Cibersortx`	Immune cell deconvolution results + boxplots
`7. Kegg_Pathview`	Pathway overlay maps: NF-κB, TCR, TGF-β
Root	PCA, dendrogram, correlation heatmap, VST expression matrix

Quality Assessment

- **PCA:** PC1 separates HT from Non-HT (primary variance axis). PC2 (28% variance) shows HRR568894 (HT) clustering between groups — a within-group outlier.
- **Sample correlations (VST):** HRR568887 ↔ HRR568888 (Non-HT): r=0.975 (tight). HRR568844 ↔ HRR568894 (HT): r=0.946 (looser). HRR568894 correlates with Non-HT samples at r=0.941–0.959.
- **CIBERSORTx deconvolution quality:** HRR568844 (HT): Correlation=0.823, RMSE=0.679 (good fit). HRR568894 (HT): Correlation=0.730, RMSE=0.725 (moderate). Non-HT samples: Correlation=0.247–0.426 (lower fit, suggesting less immune-rich profiles).

DEG Summary (Strict Thresholds: padj ≤ 0.05, $|\log_2 FC| \geq 1$)

Category	Count
Total genes tested	16,163
Upregulated in HT	**986**
Downregulated in HT	**166**
Not significant	~14,070

The strong asymmetry (986 up vs 166 down) indicates a predominantly **activatory transcriptomic program** in HT — consistent with immune infiltration and inflammatory activation.

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2. Top Differentially Expressed Genes

Thresholds applied: padj ≤ 0.05 AND $|\log_2 FC| \geq 1$. Ranked by padj (ascending), then $|\log_2 FC|$ (descending).

Top 15 Upregulated Genes in HT

Rank	Gene	$\log_2 FC$	padj	Known Biological Function
1	**CR2**	+7.95	2.26×10^{-46}	Complement receptor 2 (CD21); B cell co-receptor; binds C3d to lower B cell activation threshold
2	**LTB**	+5.58	1.70×10^{-29}	Lymphotoxin-β; drives lymphoid organogenesis and tertiary lymphoid structure (TLS) formation
3	**CCL19**	+6.58	3.54×10^{-26}	Chemokine ligand 19; recruits $CCR7^+$ T and dendritic cells; key organizer of lymphoid tissue
4	**EGR2**	+3.42	3.89×10^{-26}	Early growth response 2; transcription factor essential for Treg identity and peripheral tolerance
5	**WDFY4**	+5.82	2.40×10^{-24}	WD repeat/FYVE domain protein; cross-presentation of antigens by dendritic cells

Rank	Gene	$\log_2$FC	padj	Known Biological Function
6	**CXCR4**	+5.34	2.58×10^{-24}	CXC chemokine receptor 4; master regulator of lymphocyte homing and retention in inflamed tissue
7	**FOSB**	+3.14	1.23×10^{-21}	AP-1 transcription factor subunit; immediate-early gene activated by TCR/cytokine signaling
8	**FCMR**	+4.83	3.95×10^{-21}	Fc receptor for IgM; expressed on B cells and monocytes; modulates B cell activation
9	**ATF3**	+3.99	7.14×10^{-21}	Activating transcription factor 3; stress-responsive TF; modulates inflammatory gene expression
10	**FCRL3**	+7.19	2.57×10^{-20}	Fc receptor-like 3; B cell inhibitory receptor; FCRL3 variants are HT susceptibility loci
11	**HLA-DQB1**	+3.31	5.03×10^{-18}	MHC class II β-chain; antigen presentation to $CD4^+$ T cells; HLA-DQ alleles are primary HT risk loci
12	**RIPOR2**	+4.56	1.37×10^{-17}	Rho GTPase regulator; controls lymphocyte polarity and migration
13	**IKZF3**	+4.75	5.16×10^{-17}	Aiolos; Ikaros family TF; drives Treg and B cell differentiation programs
14	**TLR10**	+5.63	6.04×10^{-17}	Toll-like receptor 10; innate immune pattern recognition; expressed on B cells and monocytes
15	**FCRLA**	+6.19	9.57×10^{-17}	Fc receptor-like A; B cell-specific inhibitory receptor; regulates B cell activation thresholds

Top 15 Downregulated Genes in HT

Rank	Gene	$\log_2$FC	padj	Known Biological Function
1	**SLC5A5**	−2.98	2.36×10^{-18}	Sodium/iodide symporter (NIS); mediates iodide uptake into thyroid follicular cells — the primary thyroid function gene
2	**PALM3**	−3.18	1.54×10^{-8}	Paralemmin-3; membrane-associated protein involved in cell shape and process formation

Rank	Gene	$\log_2$FC	padj	Known Biological Function
3	**RAPGEF3**	−1.76	7.19×10^{-8}	Rap guanine nucleotide exchange factor 3 (EPAC1); cAMP-mediated signaling in thyroid cells
4	**NEB**	−2.66	2.42×10^{-7}	Nebulin; giant sarcomeric protein; structural role in muscle/cytoskeletal organization
5	**CHAD**	−2.23	2.63×10^{-7}	Chondroadherin; extracellular matrix protein; mediates cell-matrix interactions
6	**IGFN1**	−2.47	1.48×10^{-6}	Immunoglobulin-like and fibronectin type III domain-containing protein 1; structural/adhesion role
7	**PAX8**	−1.64	5.36×10^{-6}	Paired box 8 transcription factor; master regulator of thyroid follicular cell identity and differentiation
8	**SLC4A11**	−1.86	1.16×10^{-5}	Sodium-borate cotransporter; ion transport; expressed in thyroid epithelium
9	**SLC25A29**	−1.55	1.31×10^{-5}	Mitochondrial ornithine/citrulline carrier; amino acid metabolism in mitochondria
10	**BMP8A**	−2.46	1.72×10^{-5}	Bone morphogenetic protein 8A; TGF-β superfamily member; tissue homeostasis and differentiation
11	**RGL3**	−1.55	2.04×10^{-5}	Ral guanine nucleotide dissociation stimulator-like 3; Ras effector signaling
12	**HKDC1**	−2.13	2.78×10^{-5}	Hexokinase domain-containing protein 1; glucose metabolism
13	**EPS8L2**	−1.47	4.20×10^{-5}	EPS8-like 2; actin cytoskeleton regulation; expressed in epithelial cells
14	**RAP1GAP**	−1.49	4.77×10^{-5}	Rap1 GTPase-activating protein; negative regulator of Rap1 signaling in thyroid cells
15	**AGAP9**	−2.25	6.00×10^{-5}	ArfGAP with GTPase domain; vesicular trafficking and cytoskeletal regulation

> **Key observation:** The most significantly downregulated gene is **SLC5A5 (NIS)** — the sodium/iodide symporter that is the molecular basis of thyroid iodide uptake and the direct target of radioiodine therapy. Its downregulation in HT is

> biologically coherent and clinically significant. **PAX8**, the master transcription factor for thyroid follicular cell identity, is also downregulated, suggesting loss of thyroid epithelial cell identity under inflammatory pressure.

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3. Pathway & Functional Enrichment Analysis

3.1 GO Biological Process — Upregulated Genes (Top 5)

Rank	GO Term	Gene Count	Fold Enrichment	padj
1	Regulation of T cell activation	159	5.38	1.43×10^{-72}
2	Leukocyte cell-cell adhesion	166	5.13	1.43×10^{-72}
3	Lymphocyte differentiation	160	4.99	6.67×10^{-68}
4	Regulation of leukocyte cell-cell adhesion	153	5.19	1.31×10^{-67}
5	Positive regulation of cell activation	150	5.10	4.35×10^{-65}

3.2 GO Biological Process — Downregulated Genes (Top 5)

Rank	GO Term	Gene Count	Fold Enrichment	padj
1	Thyroid hormone generation	9	21.29	4.83×10^{-7}
2	Hormone metabolic process	22	4.97	1.27×10^{-6}
3	Thyroid hormone metabolic process	9	16.88	1.79×10^{-6}
4	Phenol-containing compound metabolic process	14	6.51	2.77×10^{-5}
5	Hydrogen peroxide biosynthetic process	5	17.00	4.39×10^{-3}

3.3 KEGG Pathways — Upregulated Genes (Top 5)

Rank	Pathway	Gene Count	Fold Enrichment	padj

Rank	Pathway	Gene Count	Fold Enrichment	padj
1	Cytokine-cytokine receptor interaction	91	3.41	8.59×10^{-25}
2	Hematopoietic cell lineage	50	5.58	2.18×10^{-24}
3	Viral protein interaction with cytokine/ receptor	48	5.36	1.56×10^{-22}
4	Intestinal immune network for IgA production	33	7.37	2.60×10^{-21}
5	Allograft rejection	29	8.30	4.34×10^{-21}

3.4 KEGG Pathways — Downregulated Genes

> ⚠️ **No KEGG pathways reach statistical significance** (padj ≤ 0.05) for the downregulated gene set after FDR correction. The top nominal hits (Cortisol synthesis, Thyroid hormone synthesis) have padj = 0.138. These are directionally consistent with the biology but should not be reported as significant enrichments.

3.5 KEGG Pathview Overlays (from uploaded files)

Three KEGG pathway maps were provided with DEG expression overlaid:

- **NF-κB Signaling Pathway:** Multiple nodes upregulated (TRAF1, BIRC3, NFKBIA, TNFSF13B, IL1B, ICAM1), consistent with inflammatory activation
- **T Cell Receptor Signaling Pathway:** Broad upregulation across TCR proximal signaling (ZAP70, ITK, LCP2, VAV1, NFATC1/2, CD40LG, CTLA4)
- **TGF-β Signaling Pathway:** Mixed regulation — some nodes upregulated (TGFB1, SMAD proteins), consistent with Treg-associated TGF-β signaling

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4. The Biological Narrative

The Core Story: Immune Takeover of the Thyroid

The transcriptomic landscape of HT vs Non-HT tissue tells a coherent and compelling story of **progressive immune infiltration displacing normal thyroid function**.

Act 1: Lymphocyte Recruitment and Tertiary Lymphoid Structure Formation

The most significantly upregulated genes — **CCL19** (+6.6 $\log_2$FC), **CXCR4** (+5.3), **LTB** (+5.6), and **CR2** (+7.9) — collectively encode the molecular machinery for lymphocyte recruitment and lymphoid tissue organization. CCL19 and CXCR4 drive chemotactic gradients that pull $CCR7^+$ T cells and dendritic cells into the thyroid parenchyma. LTB (Lymphotoxin-β) is the master organizer of tertiary lymphoid structures (TLS) — ectopic lymphoid organs that form within chronically inflamed tissues and sustain local adaptive immune responses. The co-upregulation of **WDFY4** (cross-presentation) and **HLA-DQB1** (MHC class II) indicates that antigen presentation to $CD4^+$ T cells is actively occurring within the thyroid itself.

Act 2: B Cell Activation and Autoantibody Production

The upregulation of **CR2** (CD21, B cell co-receptor), **FCMR** (IgM Fc receptor), **FCRL3** (B cell inhibitory receptor — paradoxically upregulated, possibly reflecting activated B cell subsets), **FCRLA**, and **TLR10** points to a **B cell-dominated immune infiltrate**. CIBERSORTx deconvolution confirms this: HT samples show higher naive B cell fractions (+7.8%) and memory B cell fractions (+5.3%) compared to Non-HT. This B cell expansion is the cellular basis for anti-thyroid antibody (anti-TPO, anti-Tg) production that defines HT clinically.

Act 3: T Cell Activation and Treg Expansion

The GO enrichment is dominated by T cell activation terms (padj = 10^{-72}). Key TCR pathway genes — **ZAP70**, **ITK**, **LCP2**, **VAV1**, **NFATC1/2**, **CD40LG** — are all upregulated, indicating active T cell receptor signaling. Simultaneously, **FOXP3** (+2.7 $\log_2$FC), **EGR2** (+3.4), **CTLA4** (+5.3), **IKZF3** (+4.8), and **IL2RA** (+3.9) are upregulated — a signature of **regulatory T cell (Treg) expansion**. This is a hallmark of HT: the immune system attempts to restrain the autoimmune attack through Treg expansion, but this compensatory response is insufficient to prevent thyroid damage.

Act 4: NF-κB-Driven Inflammatory Amplification

The NF-κB pathway is broadly activated: **TRAF1**, **BIRC3**, **NFKBIA** (a feedback inhibitor, upregulated as a brake), **TNFSF13B** (BAFF — B cell survival factor), **IL1B**, **ICAM1**, and **PTGS2** are all upregulated. This inflammatory amplification loop sustains the immune infiltrate and promotes tissue damage.

Act 5: Collapse of Thyroid Follicular Cell Identity

While the immune program is activated, the thyroid-specific transcriptional program collapses. **SLC5A5 (NIS)** — the sodium/iodide symporter essential for thyroid hormone synthesis — is the most significantly downregulated gene ($\log_2$FC = −2.98, padj = 2.4×10^{-18}). **PAX8** — the master transcription factor for thyroid follicular cell identity — is also downregulated ($\log_2$FC = −1.64). The GO enrichment of downregulated genes is dominated by "thyroid hormone generation" and "thyroid hormone metabolic process" (FoldEnrichment = 21.3 and 16.9, respectively). This represents the direct molecular mechanism of hypothyroidism in HT: immune-mediated suppression of the thyroid biosynthetic machinery.

Summary Model

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Immune infiltration (CCL19/CXCR4) → TLS formation (LTB) → B cell activation (CR2/FCMR/BTK)
    ↓                                                                  ↓
T cell activation (ZAP70/ITK/CD40LG) ↔ Treg expansion (FOXP3/EGR2/CTLA4)
    ↓
NF-κB amplification (TRAF1/TNFSF13B/IL1B)
    ↓
Thyroid follicular cell damage → ↓SLC5A5, ↓PAX8 → Hypothyroidism
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5. Visualizations Generated

All figures are saved in SVG (vector) and PNG (raster) formats.

Figure	File	Description
Volcano Plot	`volcano_plot_HT_vs_NonHT.svg`	DESeq2 results; top 10 up/down genes labeled; thresholds at padj=0.05,

Figure	File	Description
Heatmap	`heatmap_top50_DEGs.svg`	Top 25 up + top 25 down DEGs; Z-score normalized; samples ordered by condition
GO BP Bubble (Up)	`bubble_GO_BP_upregulated.svg`	Top 10 GO BP terms for upregulated genes; bubble size = gene count, color = significance
GO BP Bubble (Down)	`bubble_GO_BP_downregulated.svg`	Top 10 GO BP terms for downregulated genes
KEGG Barplot	`barplot_KEGG_pathways.svg`	Top 10 KEGG pathways (up only; no significant KEGG terms for down)
CIBERSORTx	`cibersortx_cell_composition.svg`	Immune cell fraction estimates per sample
Drug Targets	`drug_target_prioritization.svg`	Composite scoring of top drug candidates

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6. Actionable Insights & Next Steps

6.1 Most Promising Target/Pathway for Validation

Primary target: JAK3 / JAK-STAT signaling axis

JAK3 is upregulated 16-fold ($\log_2$FC = 4.0, padj = 4.7×10^{-8}) and sits at the convergence of TCR signaling, cytokine receptor signaling, and lymphocyte activation — the three dominant upregulated programs in this dataset. FDA-approved JAK inhibitors (tofacitinib, upadacitinib) are already in clinical use for autoimmune diseases with similar immune profiles (RA, IBD, alopecia areata), making this the most immediately translatable target.

Secondary target: CXCR4 / lymphocyte trafficking axis

CXCR4 is the most statistically significant druggable target (padj = 2.6×10^{-24}, $\log_2$FC = 5.3). Blocking CXCR4 could prevent lymphocyte homing to the thyroid, addressing the upstream recruitment step rather than downstream inflammation.

6.2 Proposed Wet-Lab Validation Experiments

Experiment 1: qPCR + Western Blot Validation Panel (Priority: Immediate)

Rationale: Validate the top DEGs in a larger cohort before any functional work.

Protocol:

- Collect thyroid tissue biopsies from ≥10 HT patients and ≥10 matched non-HT controls (euthyroid goiter or normal thyroid)
- Extract RNA → cDNA → qPCR for: **SLC5A5, PAX8, CXCR4, JAK3, EGR2, FOXP3, CR2, CCL19, LTB**
- Western blot for protein-level confirmation: JAK3, p-STAT3, CXCR4, NIS (SLC5A5)
- **Expected outcome:** Confirm directional consistency with RNA-seq; establish expression correlations with anti-TPO antibody titers and TSH levels

Experiment 2: JAK Inhibitor Functional Assay in HT-Derived Immune Cells (Priority: 3–6 months)

Rationale: Test whether JAK3 inhibition reduces the inflammatory phenotype observed in HT.

Protocol:

- Isolate peripheral blood mononuclear cells (PBMCs) from HT patients and healthy controls
- Stimulate with anti-CD3/CD28 (T cell activation) or LPS (innate activation)
- Treat with tofacitinib (JAK1/3 inhibitor) at 0.1, 1, 10 μM
- Readouts:
  - Flow cytometry: $CD4^+FOXP3^+$ Treg frequency, $CD8^+$ T cell activation markers (CD25, CD69)
  - ELISA: IL-6, IL-2, IFN-γ secretion
  - qPCR: JAK3, STAT1, CXCR4, CCL19 expression
- **Expected outcome:** JAK inhibition reduces T cell activation markers and pro-inflammatory cytokine secretion; Treg frequency may be maintained or increased

Experiment 3 (Optional): CXCR4 Blockade in a Thyroiditis Mouse Model

Rationale: Test whether CXCR4 antagonism (plerixafor) reduces lymphocyte infiltration into the thyroid.

Model: Experimental autoimmune thyroiditis (EAT) in NOD mice or CFA/thyroglobulin-immunized C57BL/6 mice **Intervention:** Plerixafor (AMD3100) 5 mg/kg s.c. daily for 4 weeks **Readouts:** Thyroid histology (H&E, lymphocyte infiltration scoring), anti-Tg antibody titers, thyroid hormone levels (T3/T4/TSH), flow cytometry of thyroid-infiltrating lymphocytes

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7. Drug Target Identification & Prioritization

7.1 Scoring Methodology

Targets were scored on a composite 100-point scale:

- **Statistical significance** (30%): normalized $-\log_{10}(\text{padj})$
- **Effect size** (20%): normalized $\log_2 FC$
- **Druggability** (30%): expert-curated score 1–5 (existing approved drugs, binding pocket availability, clinical precedent)
- **HT biological relevance** (20%): expert-curated score 1–5 (mechanistic centrality to HT pathology)

> **Transparency note:** 50% of the composite score is expert-curated (not algorithmically derived). This reflects current knowledge of druggability and disease biology and should be treated as an informed but subjective prioritization, not an objective ranking.

7.2 Top 5 Drug Targets

> **Ranking note:** Targets are listed in order of **clinical translatability priority**, which integrates the composite score with an additional consideration of existing clinical evidence in autoimmune disease. CXCR4 ranks highest by composite score (68.7) due to its exceptional statistical significance, while JAK3 (score 64.8) is elevated to the primary position because it has the strongest existing clinical precedent in autoimmunity (multiple FDA-approved indications

> with overlapping immune profiles). Where composite score and clinical translatability diverge, this is explicitly noted.

🥇 Target 1: JAK3 (Composite Score: 64.8 | Clinical Priority: #1)

Parameter	Value
$\log_2$FC (HT vs Non-HT)	+4.01
padj	4.67×10^{-8}
baseMean	810 counts
Pathways	JAK-STAT signaling, TCR signaling
Therapeutic action	**INHIBIT**

Role in HT: JAK3 is the non-receptor tyrosine kinase that transduces signals from the common γ-chain cytokine receptors (IL-2R, IL-4R, IL-7R, IL-15R, IL-21R) — all of which are upregulated in this dataset. JAK3 is expressed almost exclusively in hematopoietic cells, making it a lymphocyte-selective target. Its upregulation in HT reflects the activated state of the infiltrating T and B cell populations.

Drugs:

- **Tofacitinib** (Xeljanz) — FDA-approved for RA, psoriatic arthritis, UC, alopecia areata
- **Upadacitinib** (Rinvoq) — FDA-approved for RA, AD, AS, UC; more JAK1-selective but covers JAK3
- **Ruxolitinib** (Jakafi) — FDA-approved for myelofibrosis, PV, aGVHD

Why superior: JAK3 sits at the convergence of all three dominant upregulated programs (TCR, cytokine, lymphocyte activation). JAK inhibitors are already approved for autoimmune diseases with overlapping immune profiles. Alopecia areata — which shares HLA and immune genetics with HT — responds dramatically to JAK inhibitors, providing strong biological precedent.

🥈 Target 2: CXCR4 (Composite Score: 68.7 | Highest by composite score)

Parameter	Value
$\log_2$FC (HT vs Non-HT)	+5.34
padj	2.58×10^{-24}
baseMean	1,830 counts
Pathways	Chemokine signaling, lymphocyte trafficking
Therapeutic action	**INHIBIT**

Role in HT: CXCR4 is the most statistically significant druggable target in this dataset. It mediates lymphocyte homing to CXCL12-expressing tissues. In HT, $CXCR4^+$ lymphocytes are recruited to the thyroid where CXCL12 is produced by stromal cells. Blocking CXCR4 could prevent the initial recruitment step, addressing the disease upstream.

Drugs:

- **Plerixafor** (Mozobil) — FDA-approved for stem cell mobilization in lymphoma/myeloma

Important caveat: Plerixafor's FDA approval is for oncology (stem cell mobilization), not autoimmunity. Its use in HT would be **investigational/off-label**. No clinical trials in HT have been completed. The drug has a short half-life and is administered by injection, limiting chronic use. This target is high-priority for **preclinical validation** before clinical translation.

Why superior: Highest statistical confidence of all targets; addresses upstream lymphocyte recruitment rather than downstream inflammation; high baseMean (1,830) indicates robust expression.

🥉 Target 3: BTK (Composite Score: 59.9)

Parameter	Value
$\log_2$FC (HT vs Non-HT)	+4.45
padj	5.87×10^{-5}
baseMean	284 counts

Parameter	Value
Pathways	NF-κB signaling, B cell receptor signaling
Therapeutic action	**INHIBIT**

Role in HT: BTK (Bruton's tyrosine kinase) is the central kinase in B cell receptor (BCR) signaling and is also activated downstream of TLRs and the NF-κB pathway. In HT, B cells produce anti-thyroid antibodies (anti-TPO, anti-Tg) that mediate complement-dependent cytotoxicity and ADCC against thyroid follicular cells. BTK inhibition would block B cell activation, antibody production, and NF-κB-driven inflammatory amplification simultaneously.

Drugs:

- **Ibrutinib** (Imbruvica) — FDA-approved for B cell malignancies; off-target ITK inhibition may be beneficial in T cell contexts
- **Acalabrutinib** (Calquence) — more selective BTK inhibitor; FDA-approved for CLL/MCL
- **Zanubrutinib** (Brukinsa) — highly selective; FDA-approved for B cell malignancies

Caveat: Ibrutinib has significant off-target effects (ITK, EGFR, TEC). For autoimmune applications, more selective BTK inhibitors (acalabrutinib, zanubrutinib) or next-generation non-covalent BTK inhibitors (pirtobrutinib) are preferred. Phase II trials of BTK inhibitors in SLE and RA are ongoing.

Target 4: CTLA4 / CD28 Co-stimulation Axis (Composite Score: 66.3)

Parameter	Value
$\log_2$FC (HT vs Non-HT)	+5.28
padj	1.69×10^{-5}
baseMean	110 counts
Pathways	TCR signaling, T cell co-stimulation
Therapeutic action	**BLOCK upstream CD80/CD86 co-stimulation**

Role in HT: CTLA4 is upregulated alongside FOXP3 — both are canonical Treg markers. This co-upregulation suggests Treg expansion in HT tissue. **Important mechanistic note:** Abatacept (CTLA4-Ig) does not target CTLA4 itself — it is a fusion protein that binds CD80/CD86 on antigen-presenting cells, blocking the CD28 co-stimulatory signal required for T cell activation. CTLA4 gene variants (particularly the +49A/G polymorphism) are established HT susceptibility loci, providing strong genetic validation for this pathway.

Drugs:

- **Abatacept** (Orencia) — FDA-approved for RA and juvenile idiopathic arthritis

Critical caveat: CTLA4 is highly expressed on Tregs, which use it to suppress APCs. Abatacept could paradoxically reduce Treg suppressive capacity by blocking the CTLA4-CD80/CD86 interaction on Tregs. The net therapeutic effect in HT requires careful evaluation. This target is best pursued in the context of understanding the Treg/effector T cell balance.

Target 5: IKZF3 — Biomarker, Not Primary Drug Target (Composite Score: 62.5)

Parameter	Value
$\log_2$FC (HT vs Non-HT)	+4.75
padj	5.16×10^{-17}
baseMean	1,325 counts
Pathways	Treg differentiation, B cell programs
Therapeutic action	**BIOMARKER of disease activity**

Role in HT: IKZF3 (Aiolos) is a transcription factor that drives Treg and B cell differentiation programs. Its strong upregulation in HT reflects the expanded lymphocyte infiltrate. While lenalidomide degrades IKZF3 via the CRBN E3 ligase (used in multiple myeloma), **direct IKZF3 degradation in an autoimmune context is not recommended** — it would likely disrupt Treg function and could worsen autoimmunity. IKZF3 is better utilized as a **transcriptomic biomarker** of HT disease activity and lymphocyte infiltration severity.

Recommendation: Use IKZF3 expression as a quantitative biomarker in future studies. Indirect modulation via JAK inhibitors (which reduce the upstream cytokine signals driving IKZF3 expression) is the preferred therapeutic approach.

7.3 Additional Noteworthy Targets

Gene	$\log_2$FC	padj	Drug	Note
ITK	+3.76	3.54×10^{-9}	Ibrutinib (off-target), PRN694 (Phase I)	T cell-specific kinase; selective ITK inhibition could spare B cells
TNFSF13B (BAFF)	+3.21	6.90×10^{-4}	Belimumab (FDA-approved)	B cell survival factor; belimumab approved for SLE; rational for HT B cell pathology
IL12RB1	+4.12	5.35×10^{-5}	Ustekinumab (FDA-approved anti-IL12/23)	Th1/Th17 differentiation; ustekinumab approved for psoriasis, IBD
EGR2	+3.42	3.89×10^{-26}	No direct drug	Treg master TF; high significance but undruggable directly

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8. Critical Limitations & Discussion

8.1 Statistical Power

With n=2 per group, this study is **severely underpowered**. DESeq2 requires variance estimation across replicates; with only 2 samples per group, variance estimates are unreliable and the model relies heavily on the empirical Bayes shrinkage of dispersion estimates. The large number of significant DEGs (986 upregulated) likely reflects the strong biological signal of HT (a well-characterized disease with dramatic immune infiltration) rather than statistical robustness. However, individual gene-level estimates — particularly for lowly expressed genes (e.g., IL6, baseMean=34.6) — should be treated with caution.

8.2 Sample HRR568894 (HT Outlier)

HRR568894 shows intermediate expression patterns between HT and Non-HT groups.

This could reflect:

- **Biological heterogeneity:** HT is a spectrum; this patient may have early-stage or mild disease with less immune infiltration
- **Technical variation:** RNA quality, tissue composition, or library preparation differences
- **Recommendation:** In any follow-up study, include histological scoring of lymphocyte infiltration (Hashimoto score) to stratify HT severity

8.3 Threshold Discrepancy

The pre-existing analysis used log2FC > 1 without a padj filter (1,502 upregulated genes). The strict analysis (padj ≤ 0.05 + |log2FC| ≥ 1) yields 986 upregulated genes. The ORA enrichment results in the uploaded files were computed on the looser gene list. The pathway-level conclusions are consistent between both approaches (the dominant immune/inflammatory signal is robust), but gene-level counts differ.

8.4 Bulk RNA-seq Limitations

Bulk RNA-seq measures the average transcriptome of a mixed cell population. The HT thyroid contains thyroid follicular cells, immune infiltrate (T cells, B cells, macrophages), stromal cells, and endothelial cells. The upregulated immune signature likely reflects the **increased proportion of immune cells** in HT tissue, not necessarily transcriptional changes within individual cell types. Single-cell RNA-seq would be required to deconvolve cell-type-specific transcriptional changes.

8.5 CIBERSORTx Caveats

CIBERSORTx deconvolution with n=2 per group cannot support statistical testing. The cell fraction differences reported (e.g., B cells naive: HT=19.8% vs Non-HT=12.0%) are directionally consistent with HT biology but should not be over-interpreted. The Non-HT samples show lower CIBERSORTx correlation scores (0.25–0.43), suggesting the LM22 reference matrix may not perfectly model the Non-HT thyroid transcriptome.

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9. Conclusions

This bulk RNA-seq analysis of HT vs Non-HT thyroid tissue reveals a coherent and biologically interpretable transcriptomic signature characterized by:

1. **Massive immune activation:** 986 upregulated genes dominated by T cell activation, lymphocyte differentiation, and cytokine signaling programs
2. **B cell expansion:** CR2, FCMR, FCRL3, FCRLA, BTK upregulation; CIBERSORTx confirms increased B cell fractions in HT
3. **Treg expansion (compensatory):** FOXP3, EGR2, CTLA4, IKZF3, IL2RA co-upregulation
4. **Tertiary lymphoid structure formation:** CCL19, LTB, CXCR4 upregulation
5. **NF-κB inflammatory amplification:** TRAF1, BIRC3, TNFSF13B, IL1B, ICAM1
6. **Collapse of thyroid follicular identity:** SLC5A5 (NIS) and PAX8 downregulation — the molecular basis of hypothyroidism

Top drug targets for follow-up: JAK3 (most translatable, FDA-approved drugs), CXCR4 (highest statistical confidence, upstream target), BTK (B cell-specific, FDA-approved drugs in oncology).

All findings are hypothesis-generating and require validation in a larger, well-powered cohort with histological correlation.

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